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Effect of extreme heat stress on adverse pregnancy outcomes in India

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Supplementary material for this article is available [online](#)

Abstract

Objective. Exposure to heat during pregnancy may increase the risk of pregnancy outcomes, particularly in low- and middle-income countries (LMIC) where healthcare access and adaptive capacity are limited. Here we examined the impact of extreme heat stress in ambient conditions on adverse pregnancy outcomes in women aged 15–49 years in India from 2015 to 2020. **Methods.** We used a nationally representative population sample from India's National Family Health Survey (NFHS), a large-scale cross-sectional dataset and employed a generalised logistic mixed model to examine associations between extreme heat exposure during pregnancy and adverse outcomes - preterm birth (PTB), low birth weight (LBW), stillbirth (SB). Exposure to extreme heat was defined as the number of consecutive 'sweltering' days (characterized by the India Heat Index (IHI) exceeding the 90th percentile) throughout the gestational period and was assigned at the household level based on geocodes. Models were adjusted for potential confounders, including ambient and household air pollution, socioeconomic status, and demographic factors at both individual and cluster levels. **Results.** Among 209,266 participants, we identified 29873 LBW, 2341 PTB, and 1740 SB. Each additional day of consecutive 'sweltering' in-utero heat exposure was associated with increased risks of LBW (OR: 1.010, 95% CI: 1.001–1.018). Further, heat exposure during the trimester specific exposure window was with increased odds for PTB (OR: 1.036, 95% CI: 1.009–1.064) in first trimester, LBW (OR: 1.010, 95% CI: 1.002–1.018) in second trimester, compared to unexposed mothers. **Conclusion.** Our findings indicate that exposure to ambient heat extremes was associated with adverse pregnancy outcomes in India, and the effect varied depending on the exposure window during pregnancy. With climate change projected to increase the frequency of heat extremes, effective adaptation strategies are necessary to minimise these adverse health effects.

1. Introduction

Exposure to extreme heat events poses a significant threat to human health, particularly in lower- and middle-income countries (LMICs) like India (McElroy *et al* (2022), Maurya *et al* 2023, Yasobant *et al* 2025). Over the past decade, climate change has intensified the intensity, frequency, and duration of such events. Climate projections indicate that extreme heat events will become more frequent and intense, lasting longer (Vargas Zeppetello *et al* 2022). The rapid increase in ambient heat causes physiological strain on the body, which occurs

when it is unable to regulate its internal temperature due to excessive external heat. This condition arises when the body is exposed to more heat than it can dissipate through natural cooling mechanisms, such as sweating and increased blood circulation to the skin. This can occur in hot environments, especially when coupled with humidity, physical activity, or prolonged exposure to heat. Prolonged exposure to humid heat amplifies health risks in vulnerable populations, which has become a major public health concern.

Pregnancy is a critical period in the health of both the mother and the foetus, marked by numerous physiological changes that occur within the mother. The metabolic process of the foetus generates heat, which is naturally dissipated through the placenta. As part of this process, the foetal-placenta unit demands maternal cardiac output to support the foetus (Ramirez *et al* (2024)). To accommodate this, maternal cardiac output increases by 30 to 50% toward the end of the pregnancy, putting additional stress on the maternal cardiovascular system. Changes occur in the maternal core body temperature (Soma-Pillay *et al* 2016). In this context, exposure to extreme heat may further elevate core temperature, potentially double the risk of adverse pregnancy outcomes and negatively impact both maternal and foetal health. Some research evidence shows the potential biological effects of high ambient temperature: (1) Increased maternal core body temperature raises foetal core temperature, leading to birth defects. (2) Dehydration decreases uterine blood flow and foetal growth, and due to reduced blood flow, there is a heightened chance of uterine contractility and an increased risk of preterm birth (Barreca and Schaller 2020, Samuels *et al* 2022, Bansal *et al* 2023, Bonell *et al* 2020 2023). LMICs experience higher birth rates and a greater prevalence of adverse birth outcomes (Hug *et al* 2021).

Growing evidence suggests that continued exposure to ambient heat has a negative influence on foetal growth and pregnancy outcomes in India (Rekha *et al* 2024). Exposure to high ambient temperatures can increase the risk of premature birth (before 37 complete weeks of gestation), low birth weights (live births under 2500 g), stillbirths (foetus die after 28 weeks), neonatal deaths (death occurring 0–28 days after birth) and miscarriage (foetus die before 28th week). Limited studies have explored the impact of heat stress on pregnancy outcomes in India (Kakkad *et al* 2014, Rekha *et al* 2024). A major limiting factor is the lack of indigenous evidence regarding the health impacts of extreme heat stress in India, along with the difficulty in attributing exposure at the population scale due to insufficient systematic exposure data and a lack of a framework to integrate it with health data.

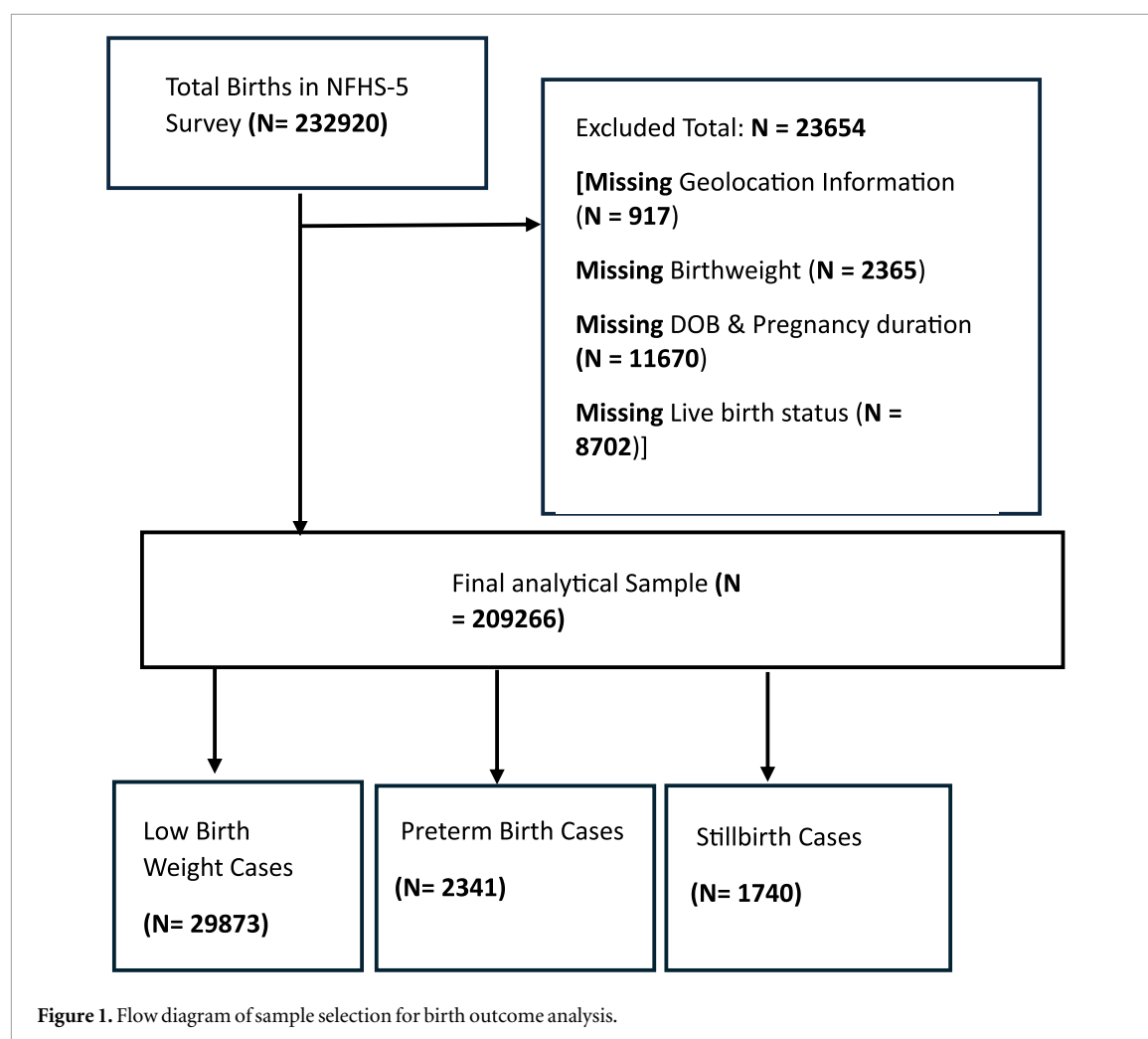
Understanding the timing and magnitude of exposure is also crucial, particularly in identifying critical gestational windows during which heat stress has the greatest impact on maternal and foetal health (Hajdu and Hajdu (2021)). Traditional temperature-based exposure indicators, which rely solely on air temperature, fail to capture the full complexity of heat stress, as it is influenced by factors such as humidity, wind speed, and solar radiation (Zhang *et al* 2014, Heidari *et al* 2020, Pantavou *et al* (2025)). Findings suggest that extreme heat stress events trigger adverse physiological responses during pregnancy compared to average exposure (Yang *et al* (2022), Hanson *et al* 2024, Lakhoo *et al* 2025). Rather than relying on average or maximum exposure metrics during the exposure windows, our study focused on counting extreme heat stress days to better capture extreme heat stress events.

To fill this critical knowledge gap, we leveraged nationally representative data from the fifth round of the National Family Health Survey (NFHS-5) and the India Heat Index (IHI), a modified measure of wet bulb globe temperature (Dash *et al* 2017, Dey *et al* 2021, Choudhary *et al* 2023), as exposure to extreme heat stress. We examined and report (1) how pregnancy outcomes were affected by exposure to consecutive days with the IHI exceeding the 90th percentile of the long-term local distribution during the pregnancy period, (2) the effect modification of this association by potential risk factors such as the foetus's gender, the mother's body mass index (BMI), mother's age, education, and other socio-demographic covariates like residence area type, wealth index, conception season and air pollution, (3) how the effect estimates varied across different climate zones according to Koppen climate classification in India through subgroup analysis, and (4) the robustness of the statistical model through sensitivity analysis, assessing whether the association changed with an increase in the length of consecutive extreme heat stress days.

2. Data and methods

2.1. Study population

The Demographic Health Survey (DHS) is a nationally representative household survey that assesses maternal and child health indicators in India, referred to as the National Family Health Survey (NFHS-5, 2019–2021). It collects data from women aged 15–49 across selected households in each cluster within 707 districts covering 29 states and six union territories of India. This survey offers information on the household and individual socio-demographic characteristics. The survey employs a two-stage sampling method, stratified by rural and urban areas. In rural regions, the primary sampling unit is a village segment, whereas in urban areas, it is a census enumeration block, collectively referred to as a cluster. We obtained all the information about covariates



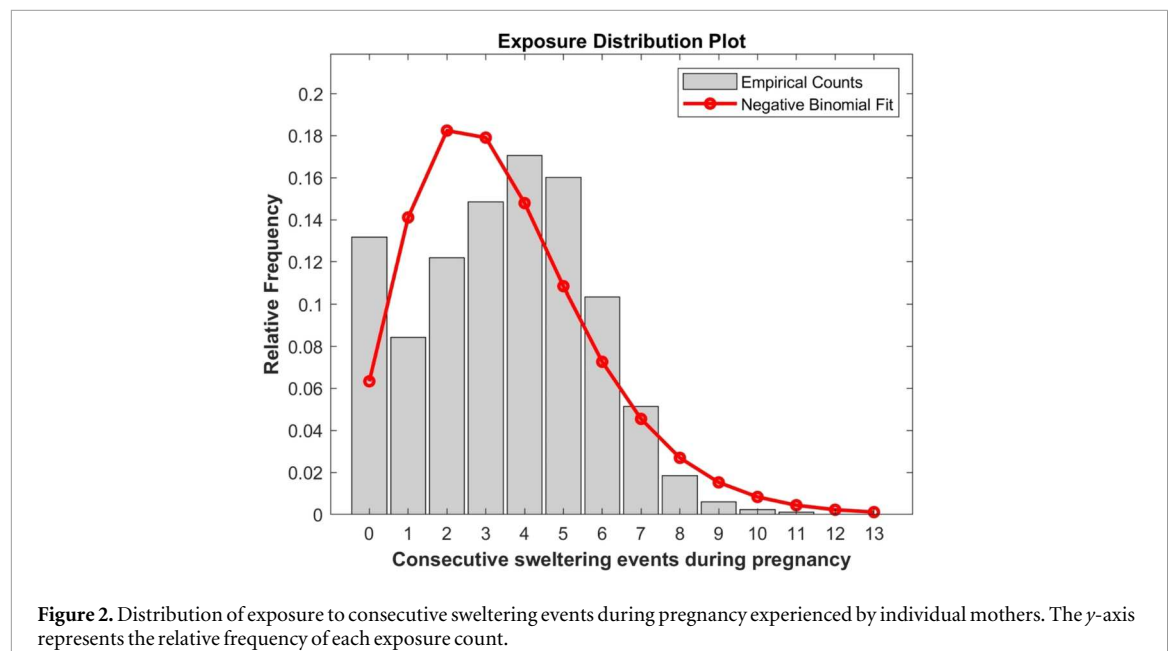
(anaemia level, Mother's age, BMI, Education, Smoking status, Wealth index, Residence type) from the NFHS-5 questionnaires administered to 258434 women ages 15–49 (IR file) and their children (BR file), For each eligible woman's children under five, the survey records a comprehensive birth and death history, including details such as date of birth, gender, pregnancy duration, birth weight, and live birth status. There were 232,920 births recorded in the original NFHS-5 dataset of which 23654 children had missing geolocation information, birth weights, birth date, pregnancy duration, and birth history (figure 1).

2.2. Outcome classification

Preterm birth status was identified using the 'pregnancy duration' variable from the birth history questionnaire. Births occurring before 37 weeks of gestation were classified as preterm birth. Miscarriage, Stillbirth and abortion were obtained from the 'Pregnancy end in miscarriage, Stillbirth and abortion' variable. However, abortions were not included in this analysis because they are predominantly induced rather than spontaneous events, and existing literature does not establish a consistent causal association between heat exposure and induced abortions. We excluded Miscarriages because of the difficulty in interpreting third-trimester self-reported miscarriage as a coherent clinical outcome. The stillbirths verified based on responses to the questions 'Is the child alive or dead at the time of interview?' and 'Age at death.' If the child was reported as not alive at the time of interview and the age at death indicated death at birth, the case was classified as a stillbirth. Low birth weight was defined as a reported 'birth weight' of less than 2500 g.

2.3. Confounders and effect modifiers

Potential risk factors for PTB, LBW, and SB were extracted from the dataset after examining recent literature (Kuppusamy *et al* 2023, Ravanelli *et al* 2026, Soma-Pillay *et al* 2016, Rekha *et al* 2024). The extracted variables were categorized into individual, household, and cluster levels. Variables related to maternal and child characteristics were specific to each woman and child, classified as individual-level variables—sex of the child, mother's education, mother's body mass index, mother's age, anaemic level, and birth order. Smoking status



was also recorded in this survey, where an individual smoking bidi, cigarettes, or tobacco was identified as a smoker. Detailed descriptions of covariates are presented in **table S1**.

Household-level variables included characteristics applicable to all mothers residing in the same household, such as the wealth index and cooking fuel type. The cluster-level variables pertained to all women and children living in the same cluster, including their place of residence and climate zone. The climate zone of each cluster was identified using the provided geolocation information. Socio-economic status was calculated using the individual household's wealth index, categorized into 'poorest', 'poorer', 'middle', 'richer', and 'richest'. The wealth index categories are defined based on the range of a household's cumulative living standards, which include household assets and amenities.

Ambient and household air pollution are associated with pregnancy outcomes in India (Balakrishnan *et al* 2018, Goyal and Canning 2021, Parchure *et al* (2024)) and are also considered in this study. To address this, we included the cooking fuel types - Electricity, LPG, Natural gas, biogas, kerosene, wood, coal, animal dung - categorized into unclean and clean cooking fuel in the survey data. For ambient air pollution exposure, we extracted satellite-derived daily ambient PM_{2.5} that has been developed and evaluated for India (Katoch *et al* 2023) using the participants' cluster geolocation.

2.4. Heat exposure attribution

The India heat index (IHI) is a crucial metric for assessing the combined effects of multiple meteorological variables on human heat exposure (Dash *et al* 2017, Choudhary *et al* 2023). This index provides a more comprehensive exposure measure by integrating meteorological variables to assess heat stress.

The National Centre for Medium-Range Weather Forecasting (NCMRWF) at the Centre of Excellence in Weather and Climate Modelling under the Ministry of Earth Sciences provides gridded data of on-air temperature, relative humidity above 2 meters from the surface, windspeed and downward shortwave solar radiation flux at a spatial resolution of 0.1° (~10 km), covering all of India. Downloaded the above meteorological parameters on an hourly basis and validated these variables with ground observational data (Supplementary figure S3). Finally, we calculated hourly IHI from 2015 to 2020, using the empirical equation (equation (1)) using MATLAB R2024b software.

$$\text{India Heat Index} = 0.865 \times \text{Air Temperature} + 0.146 \times \text{Relative humidity} - 0.168 \times \text{Wind speed} + 0.003 \times \text{Solarradiation} - 8.209 \quad (1)$$

The processed hourly IHI was averaged to compute the daily mean IHI, which was assigned to each primary sampling unit (rural and urban clusters) based on their geocode (Supplementary figure 1) and matched to 232920 mother-infant pairs (figure 2). Estimated cluster-wise 90th, 75th, 50th, and 25th percentiles of heat stresses for the study period for each participant. The exposure windows (Entire pregnancy period in days, Trimester 1, Trimester 2, Trimester 3) are calculated for each pregnant woman, from the recorded information regarding the date of birth (DOB) and date of conception (DOC) for each pregnant woman and their children from the survey. Counted heat extreme days - 'Very hot' days (>75th to ≤90th) and 'Sweltering' days (>90th), based on the percentile method described in Dey *et al* (2021), during the pregnancy windows of each individual.

To determine the potential exposure indicators for extreme heat, we calculated number consecutive very hot days (≥ 3 days with IHI > 75 th to ≤ 90 th percentile), consecutive sweltering days (≥ 3 days with IHI > 90 th percentile), along with very hot days and sweltering days of exposure to the mother during her pregnancy and repeated the model analysis.

Each daily IHI value measure was linked with daily individual cases of preterm births, low birth weights and stillbirths using information about the geographic coordinates of survey participants and timing of birth.

2.5. Statistical analysis

We performed an extended generalised linear mixed model (GLMM) with logit function to estimate the impacts of extreme heat days on adverse pregnancy outcomes (Chaudhary *et al* 2023). Statistical analysis was done using R v.4.5.1 version. Both univariable and multivariable analyses were conducted by applying individual-level weights in separate models for each pregnancy outcome. Multivariable analyses undertaken included the individual-level, household-level, and cluster-level covariates: Mother's BMI, maternal education, foetus sex, maternal age, birth order, anaemia level, smoking status, wealth index, residence type, cooking fuel type, climate zone, and ambient PM_{2.5} the mother belongs to. The potential covariates were identified from the previous epidemiological studies (Lakhani *et al* 2024, Rekha *et al* 2024). A stratified analysis was conducted using the interaction model to assess the effect modification (figure 5) of the association between consecutive sweltering days and pregnancy outcomes. Stratified by residence, wealth index, mother's education, infant sex, smoking status, birth order, mother's age, mother's BMI. Subgroup analysis was used to calculate the impact of changes in various climate zone classes, according to the Koppen climate classification in India (figure S4). We conducted sensitivity analysis, to assess whether the observed exposure-outcome relationship was changing with increasing number of days in continuous sweltering event.

3. Results

3.1. Descriptive analysis

Between 2015 and 2020, 232,920 births were recorded across 707 districts and 28 states in India, according to NFHS-5. In the analytical sample (figure 1), there were 166,001 births recorded in rural (villages) and 43,265 births in urban (enumeration block) clusters. After excluding 23,654 births from the study due to missing cluster geolocation information, delivery date, birth weight, pregnancy duration (in months), and live birth status, the final sample size included 209,266 births used in the analysis (table 1).

The national-level prevalence rates calculated were 12.8% for LBW, 1.005% for PTB, 0.74% for SB from the total analytical sample of 209,266 births. Figure 3 illustrates the spatial distribution of district-wise prevalence in India for each pregnancy outcome, as reported in the survey. The prevalence was slightly higher in female foetuses (51.1%) than in males (48.9%) for LBW. For PTB, the rates were 53.4% in male foetuses and 46.6% in females. For SB, the prevalence was identified as 50% in both females and males. The detailed demographic characteristics of the sample size and pregnancy outcomes are provided in table 1.

3.2. Risk of adverse pregnancy outcomes associated with extreme heat

Figure 4 illustrates the relationship between prolonged extreme heat (consecutive sweltering days) and adverse birth outcomes based on the odds ratios (OR) derived from the statistical model. For each additional day of continuous sweltering exposure, the odds of low birth weight (LBW) increased by 1% (OR = 1.012; 95% CI: 1.003–1.020). The corresponding odds ratios for Stillbirth (SB) were 1.105 (95% CI: 0.571–2.137), in the unadjusted models. After adjusting for confounders, the adjusted ORs of LBW were found to be 1.010 (1.001–1.018) (Table S2), for every additional day increase in continuous sweltering days (figure 4). In contrast, PTB showed a negative association with in utero - specific exposure in both unadjusted and adjusted models.

We found, exposure to extreme heat during the first trimester was identified as critical for PTB (aOR: 1.036, 1.009–1.064), while similar exposure in the second trimester was critical for LBW (aOR: 1.010, 1.002–1.018). The association between SB and extreme heat exposure higher in 3rd trimester for SB (aOR: 1.029, 0.998–1.060) compared to first and second with a wide confidence interval.

3.3. Effect modification and subgroup analysis

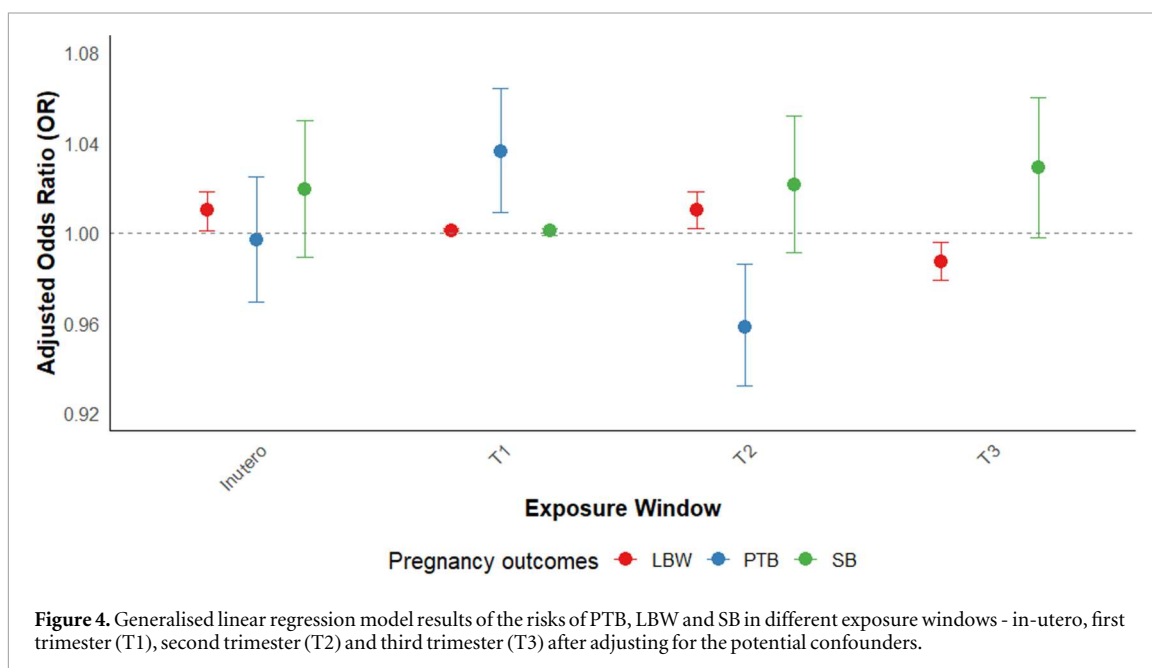
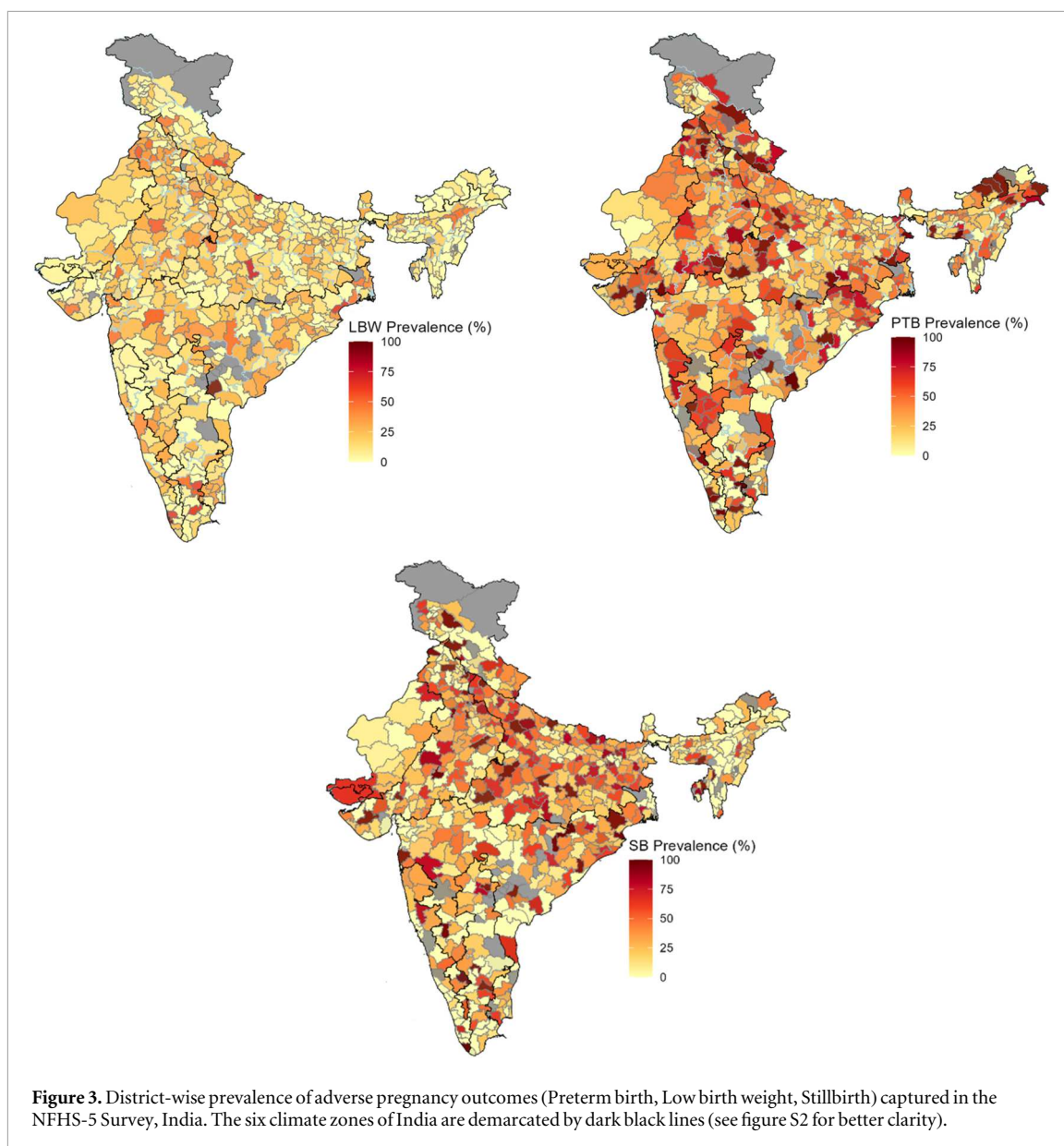
Each pregnancy outcome models were further analysed for effect modification using potential effect modifiers (figure 5). The results indicated that the risk of extreme heat exposure was modified differently across various pregnancy outcomes. The results suggested some variation in estimates across different subgroups. For stillbirth (SB), The estimated association with extreme heat exposure was slightly higher among female foetuses than male foetuses (1.05, 95% CI: 1.00–1.07 vs. 1.03, 95% CI 1.03–1.08), the confidence intervals substantially

Table 1. Characteristics of reported birth outcomes (Low birth weight, Preterm birth and Stillbirth) and the prevalence during the study period.

	LBW (N = 29873) count	LBW pre- valence (12.8%) %	PTB (N = 2341) count	PTB Pre- valence (1.005%) %	SB (N = 1740) count	SB Pre- valence (0.74%) %
Infant sex						
Male	14608	48.9	1251	53.44	870	50
Female	15265	51.1	1090	46.56	870	50
Birth Order						
1	22267	74.54	1565	66.85	1363	78.33
2	6790	22.73	645	27.55	335	19.25
3	776	2.6	118	5.04	40	2.3
4	40	0.13	13	0.56	1	0.06
Anaemia level						
Severe	796	2.66	74	3.16	57	3.28
Moderate	9247	30.95	687	29.35	564	32.41
Mild	7450	24.94	564	24.09	430	24.71
Not anaemic	11227	37.58	925	39.51	614	35.29
NA	1153	3.86	91	3.89	75	4.31
Mother's Age group						
15–19	933	3.12	98	4.19	19	1.09
20–34	26324	88.12	1994	85.18	1603	92.13
>35	2616	8.76	249	10.64	118	6.78
Mother's BMI						
<18.5	6695	22.41	466	19.91	315	18.1
18.5–24.9	17577	58.84	1345	57.45	1049	60.29
>25	4745	15.88	468	19.99	324	18.62
NA	856	2.87	62	2.65	52	2.99
Mother's education						
No education	6641	22.23	440	18.8	446	25.63
Primary	4112	13.76	293	12.52	285	16.38
Secondary	15577	52.14	1323	56.51	845	48.56
Higher	3543	11.86	285	12.17	164	9.43
Smoking status						
No	29703	99.43	2328	99.44	1734	99.66
Yes	170	0.57	13	0.56	6	0.34
Wealth index						
Poorest	8310	27.82	588	25.12	586	33.68
Poorer	7151	23.94	551	23.54	475	27.3
Middle	5671	18.98	497	21.23	306	17.59
Richer	4963	16.61	400	17.09	231	13.28
Richest	3778	12.65	305	13.03	142	8.16
Residence Type						
Urban	6007		490		251	14.43
Rural	23866		1851		1489	85.57
Climate zone						
Arid	743	2.49	42	1.79	20	1.15
Humid subtropical	11024	36.9	795	33.96	944	54.25
Mountainous	1607	5.38	217	9.27	83	4.77
Semi-arid	8782	29.4	726	31.01	408	23.45
Tropical wet	1037	3.47	70	2.99	26	1.49
Wet and dry	6553	21.94	483	20.63	257	14.77
NA	127	0.43	8	0.34	2	0.11

overlapped, indicating no clear evidence of a statistically significant difference between sexes. A similar pattern observed for low birth weight (LBW) (1.02, 95% CI 1.02–1.03 versus 0.99, 95% CI 0.98–1.0) model.

BMI (kg/m^2) was another effect modifier as the association between heat extremes and LBW and SB outcomes varied across different BMI groups. Mothers who had a low ($\text{BMI} < 18.5$) BMI showed a higher risk of LBW (aOR: 1.05, 1.04–1.06) compared to those with normal ($18.5\text{--}25.9 \text{ kg m}^{-2}$) BMI (aOR: 1.00, 0.991–1.004) for each additional day of exposure to heat extremes (figure 5(a)). Among smoking mothers, the association



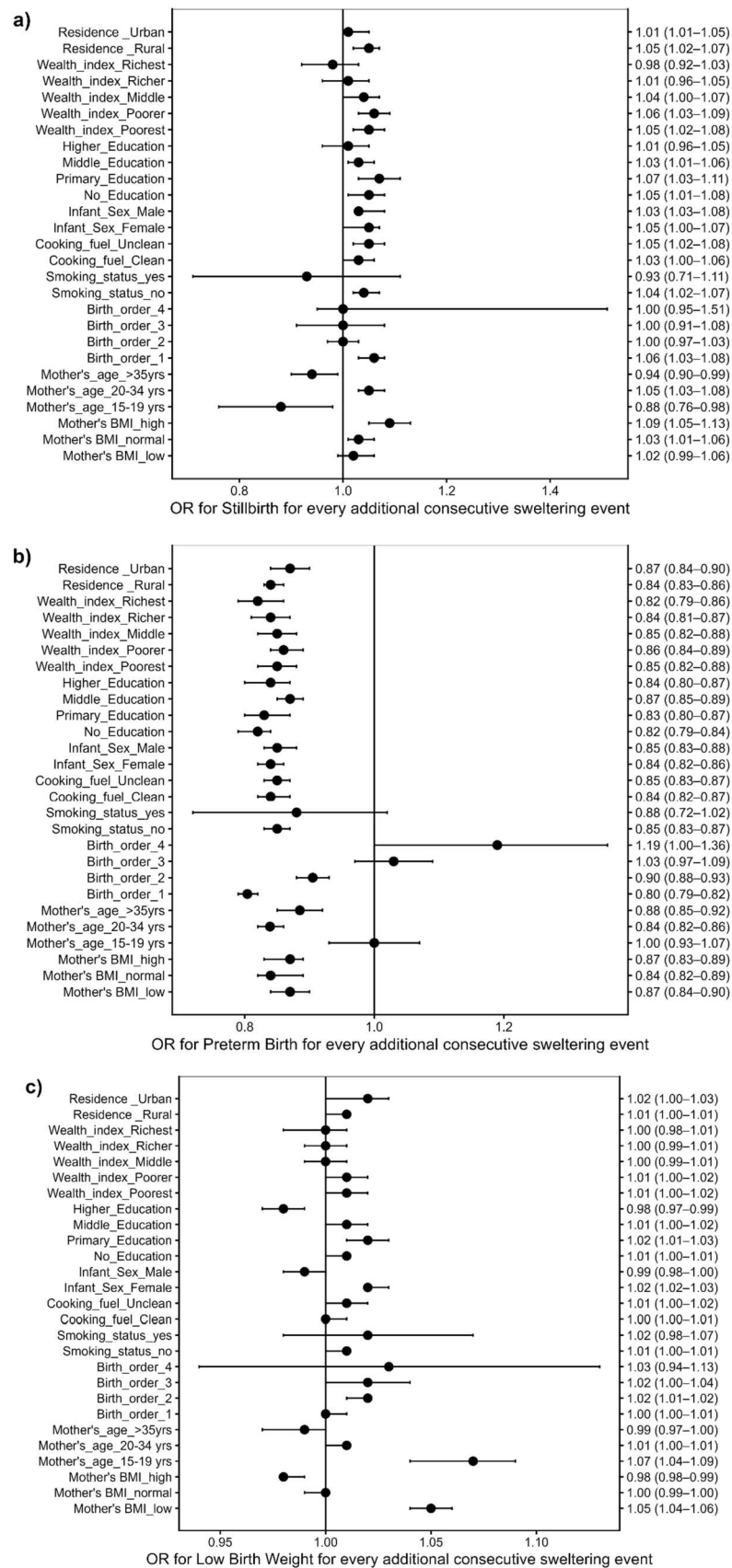


Figure 5. Effect modification of consecutive sweltering events on adverse pregnancy outcomes using interaction models. Panel (a) represents the odds ratio (OR) and 95% confidence intervals (CI) for stillbirth (SB) associated with each additional consecutive sweltering event during pregnancy. Panels (b), (c) display corresponding associations for preterm birth (PTB), and low birth weight (LBW) respectively.

Table 2. Adjusted ORs of pregnancy outcomes using the number of days of consecutive sweltering events (* implies $p < 0.05$) during pregnancy period.

Number of days in consecutive sweltering events during pregnancy duration	Low birth weight	Preterm birth	Stillbirth
3–4 days	1.044(1.002–1.087) *	1.011(0.884–1.155)	1.045(0.895–1.221)
5–6 days	1.020(0.976–1.065)	0.890(0.768–1.032)	0.978(0.827–1.157)
7+ days	1.006(0.946–1.070)	0.996(0.811–1.224)	1.063(0.844–1.338)

between consecutive sweltering days and LBW was positive but not statistically significant (aOR: 1.02, 95% CI: 0.986–1.075). Among non-smoking mothers, a small but statistically significant association was observed (aOR: 1.01, 95% CI: 1.005–1.018). Positive associations were observed in both urban and rural residents. The association was slightly higher among mothers residing in urban areas (aOR: 1.02; 95% CI: 1.007–1.027) compared with rural areas (aOR: 1.01; 95% CI: 1.004–1.017). Risk of Stillbirth increases from ‘Richest’ to ‘Poorest’ wealth index categories. Mothers with ‘no education’ and only ‘primary education’ were more likely to give birth to babies with LBW and to suffer from SB compared to mothers with higher education for an increase in continuous sweltering days. The PTB interaction model results present no positive association between most of the modifiers (figure 5(b)).

3.4. Sensitivity analysis

Exposure to consecutive sweltering days during the pregnancy duration from the date of conception to delivery was associated with adverse pregnancy outcomes in almost all climate zones (figure S4). We found that mothers living in mountainous climate zones were more affected by the continuous heat extremes than those living in other regions, probably because they were less adaptive to extreme conditions. Apart from mountainous region, LBW were significantly associated with exposure to continuous heat extremes in humid subtropical and arid region. In other climate zones, most of the pregnancy outcomes were positively associated, but were not significant, most likely, due to a low number of overall sample size for such extremes.

Further sensitivity analysis (table 2) found that the impact of consecutive sweltering days on adverse pregnancy outcomes varied with the number of days (duration) in the consecutive events during the entire pregnancy period (3–4 days, 5–6 days, and ≥ 7 days). For low birth weight (LBW), exposure to 3–4 consecutive sweltering days was associated with a modest increase in odds (OR: 1.044; 95% CI: 1.002–1.087) with a statistical significance. In contrast, the estimates for 5–6 days (OR: 1.020; 95% CI: 0.976–1.065) and ≥ 7 days (OR: 1.006; 95% CI: 0.946–1.070) were close to the null and not statistically significant. For preterm birth (PTB) and Stillbirth, none of the exposure categories showed a statistically significant association.

4. Discussion and conclusions

Exposure to heat extremes poses a major health risk, particularly for vulnerable groups such as pregnant mothers. However, studies examining the impact of extreme heat on pregnancy outcomes are scarce in India (and other LMICs) due to insufficient ground-monitoring sites, resulting in inadequate exposure attribution and limited cohort data with regular follow-ups. While a new multi-centric study, ‘Heat in Pregnancy’ (HiP), has been launched in India (<https://wellcome.org/research-funding/funding-portfolio/funded-grants/effects-extreme-heat-maternal-placental-and-fetal>), we present the first population-scale evidence of the association between prolonged extreme heat exposure and pregnancy outcomes, utilising data from the Demographic Health Survey. A key strength of our study is the extensive geographical coverage of the nationally representative survey dataset, which facilitates the evaluation of associations across India and enables subgroup analysis. High-resolution gridded exposure data helped us to triangulate exposure at the household level, thereby reducing the likelihood of exposure misclassification.

However, our study also has limitations. First, some of the outcomes, such as SB prevalence, were derived from self-reported data, which can introduce some uncertainty. Secondly, some of the important confounders - gestational weight, gestational diabetes, pre-eclampsia, hypertension, occupational exposure and physical activity during pregnancy (Bonell *et al* 2022, Rekha *et al* 2024) could not be considered in the analysis because of the unavailability of such information in the health survey data and their omission can create residual confounding. Our sample included births that occurred within the five years preceding the survey among women of reproductive age (15–49 years) residing in the sampled households. We could not consider children whose mothers were deceased or absent during the survey. Finally, ambient heat condition was used as a proxy for individual exposure, which may not fully capture personal variations in heat extremes. Alternative heat exposure metrics may better capture acute physiological stress. Sensitivity analysis using consecutive sweltering days

defined by IHI maximum showed null associations relative to current IHI -mean based metric. We considered that participants from the same cluster experienced the same amount of exposure. The bias due to the cluster effect was corrected in the statistical analysis by giving the cluster number as a separate variable in the logistic regression model.

Previous epidemiological studies have explored the association between heat exposure and pregnancy outcomes worldwide (Doke *et al* 2021, Kuppusamy *et al* 2023, Shankar *et al* 2023, Rancière *et al* 2024, Ai *et al* 2025). Furthermore, cohort-based studies have provided significant evidence of the association between heat and birth outcomes. Bonell *et al* (2024) hypothesised a physiological pathway that explains the impact of heat on the placenta and foetus. Wu *et al* (2023) conducted an extensive population-based cohort study with 205,771 participants, including 18,014 PTB cases from 2010 to 2018. Their findings indicated that exposure to temperature at the 95th percentile compared to the median in the week before birth was associated with a hazard ratio of 1.36 (1.31–1.41), even after adjusting for maternal factors such as BMI, anaemia, smoking, alcohol consumption during pregnancy, and offspring sex. Similarly, Wang *et al* (2024) analysed 10,826 mother-child pairs from 2013 to 2018, demonstrating an increased OR of 1.39 (0.87–2.20) for PTB at the 95th percentile of temperature compared to the median in the week before birth. This study accounted for maternal age, BMI, parity, season of conception, smoking or alcohol use during pregnancy, regional variations, and relative humidity. In South Asia, Rekha *et al* (2024) conducted a prospective cohort study involving 800 pregnant women engaged in physically demanding work and found that women exposed to high heat levels had a twofold increased risk of miscarriage. Our study explored the extreme conditions, and more importantly, the effect of prolonged extreme conditions as the key exposure metric.

Most studies investigating the impact of heat exposure have used maximum and minimum temperatures as metrics but have not considered the effect of exposure to prolonged and intense heat stress. Use of the IHI in this study ensured that other meteorological factors such as humidity, wind and radiation were also considered. The thresholds to identify ‘sweltering’ conditions, based on long-term climate statistics, capture the heterogeneity across climate zones (Chaudhary *et al* 2023) and are therefore more suitable for Indian conditions. We selected LBW, PTB, SB as the pregnancy outcomes, as they are the most common reported outcomes in India (Tandon *et al* 2023).

A key finding was that when high heat stress occurs continuously at varying thresholds and durations, the adverse effects on lbw differ significantly. Trimester-wise analysis helped to find the potential critical window in the pregnancy period for each pregnancy outcome. The stratified analysis from our study confirmed that pregnancy risks vary across India’s climate zones. Also, the risk was higher for economically low and high sub-groups. The sensitivity analysis helped to identify potential biases and assess the model’s robustness.

Our study emphasizes the heightened risk of adverse pregnancy outcomes in India due to prolonged extreme conditions. Such extreme conditions are expected to increase due to global warming; therefore, India should prepare to address them with more adequate adaptation planning. It is crucial to raise awareness of the risks of heat exposure for pregnant women from marginalized populations, particularly those with lower education and residents of rural districts. Finally, we call for cohort studies to further validate these associations and understand biological causal pathways using additional data on biomarkers collected during the pregnancy period.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Ethical statement

The current study was assessed as exempt from ethics review by the University of Queensland Health Research Ethics Committee (No: 2024/HE001150).

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